THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angles of a quadrilateral are in the ratio 3:4:5:8. The difference between the largest and the smallest angle is: 1

(a) 198° **(b)** 102° **(c)** 144° **(d)** 90°

2. One angle of a parallelogram is 2 times its adjacent angle. The smaller angle is: 1

(a) 60° **(b)** 120° **(c)** 80° **(d)** 90°

3. In a rhombus ABCD, $\angle ACD = 25^\circ$. Then $\angle ABC$ is: 1

(a) 65° **(b)** 50° **(c)** 130° **(d)** 155°

4. The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 100^\circ$, then $\angle OAB$ is: 1

(a) 40° **(b)** 50° **(c)** 65° **(d)** 100°

5. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = (3x + 10)^\circ$ and $\angle D = (4x + 30)^\circ$. The measure of $\angle C$ is: 1

(a) 130° **(b)** 70° **(c)** 110° **(d)** 145°

6. The side of a rhombus is $(3x + 2)$ cm and its perimeter is 80 cm. The value of x is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

7. A quadrilateral has equal diagonals that do not bisect each other. It can be an: 1

(a) rhombus **(b)** kite
(c) parallelogram **(d)** isosceles trapezium

8. The angles of a quadrilateral are in the ratio 1:3:3:5. The difference between the largest and the smallest angle is: 1

(a) 180° **(b)** 150° **(c)** 132° **(d)** 120°

9. **Assertion (A):** The adjacent angles of a parallelogram are supplementary. 1
Reason (R): The opposite sides of a parallelogram are parallel, so adjacent angles are co-interior angles on a transversal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The diagonals of a rectangle are equal in length. 1
Reason (R): The diagonals of every parallelogram bisect each other.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In a parallelogram ABCD, $\angle A = (4x + 14)^\circ$ and $\angle C = (5x + 2)^\circ$. Find x and $\angle B$. 2

12. The angles of a quadrilateral are in the ratio 2:5:6:7. Find the largest angle and say whether it is acute, right or obtuse. 2

13. In a kite ABCD, AB = AD, CB = CD, $\angle A = 70^\circ$ and $\angle B = 100^\circ$. Find $\angle C$ and $\angle D$. 2

14. In a parallelogram ABCD, $\angle A = (3x + 30)^\circ$ and $\angle C = (5x + 10)^\circ$. Find x and $\angle B$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a parallelogram ABCD with $\angle A = 50^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

16. The angles of a quadrilateral are x°, $(2x - 10)^\circ$, $(3x - 20)^\circ$ and $(4x - 30)^\circ$. Find all four angles. 3

17. One angle of a parallelogram is 2 times its adjacent angle. Find all the angles of the parallelogram. 3

OR

In a parallelogram ABCD with $\angle A = 70^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

18. The angles of a quadrilateral are x°, $(2x - 20)^\circ$, $(3x - 40)^\circ$ and $(4x - 60)^\circ$. Find all four angles. 3

OR

One angle of a parallelogram is 3 times its adjacent angle. Find all the angles of the parallelogram. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A carpenter makes a parallelogram-shaped window frame ABCD with $\angle A = 70^\circ$, $AB = 80$ cm and $BC = 50$ cm. The diagonals AC and BD meet at O, with $AC = 96$ cm and $BD = 60$ cm. 5
(i) Find $\angle C$.
(ii) Find $\angle B$.
(iii) Find the perimeter of the frame.
(iv) Find OA and OB, and state the property used.

20. In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 2 : 3 : 3 : 4$. Find the four angles. Show that AB $\parallel$ DC. 5

OR

In a parallelogram ABCD, AB $= (4x + 2)$ cm, CD $= (3x + 9)$ cm and AD = 15 cm. Find x, the lengths of AB and CD, and the perimeter. 5

END OF QUESTION PAPER — **Total: 40 Marks**